

Wave motion — a fixed string

An undamped string released from its first mode with zero initial velocity.

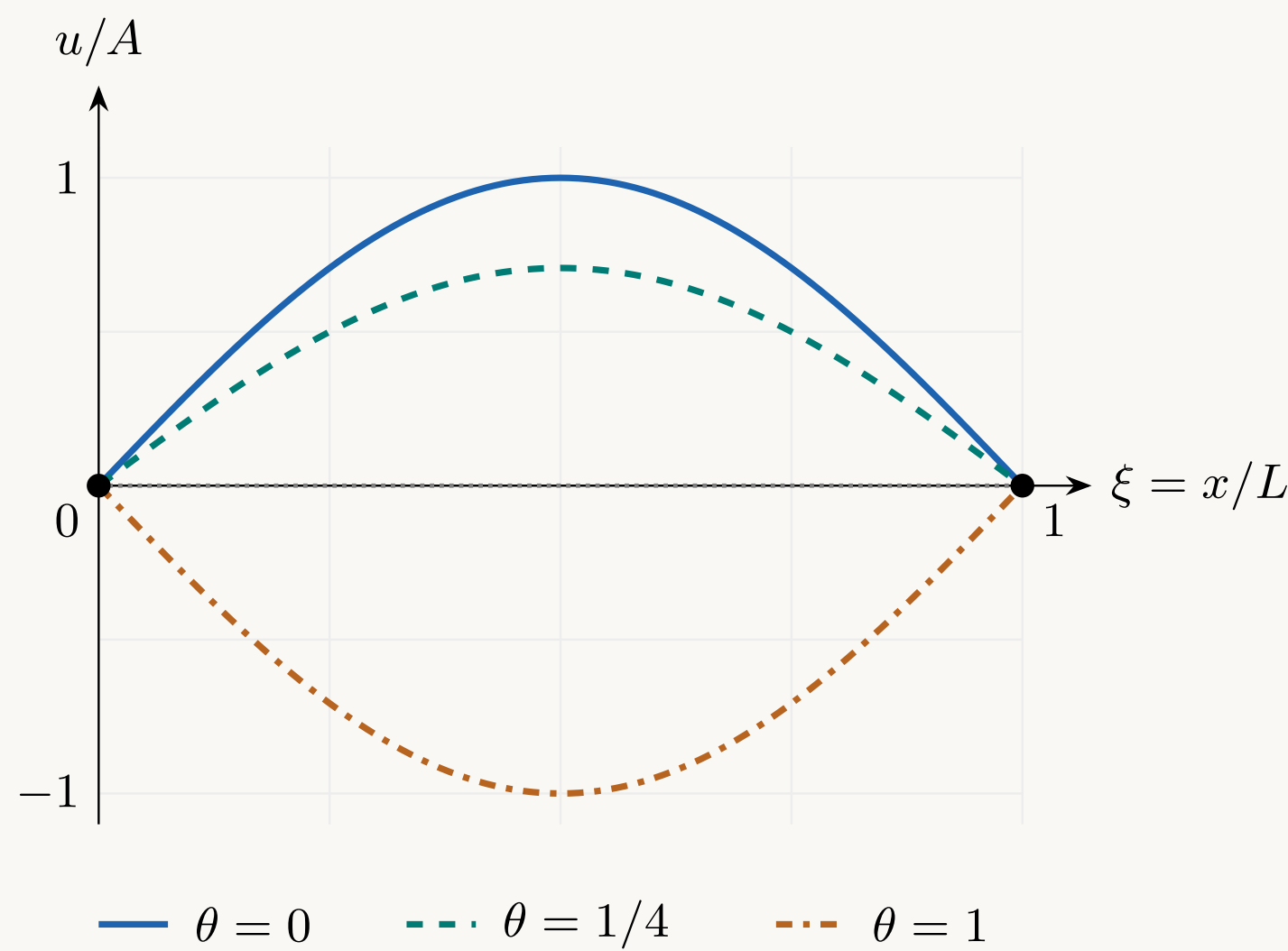
01 / MODEL + CONDITIONS

$$u_{tt} = c^2 u_{xx}, \qquad 0 < x < L, \quad t > 0,$$
$$u(0, t) = u(L, t) = 0, \qquad c > 0,$$
$$u(x, 0) = A \sin(\pi x / L), \qquad u_t(x, 0) = 0,$$
$$\xi = x / L, \qquad \theta = ct / L, \qquad v = u / A,$$
$$v_{\theta\theta} = v_{\xi\xi}.$$

02 / EXACT SOLUTION + CHECK

$$\omega_1 = \pi c / L, \qquad T = 2L / c,$$
$$u(x, t) = A \cos(\omega_1 t) \sin(\pi x / L),$$
$$E(t) = \tfrac{1}{2} \int_0^L [u_t^2 + c^2 u_x^2] \, dx,$$
$$E'(t) = c^2 [u_t u_x]_0^L = 0,$$
$$E(t) = \frac{A^2 c^2 \pi^2}{4L}.$$

03 / GEOMETRY + BEHAVIOUR



At dotted $\theta = 1/2$: $u = 0$, speed is largest; energy is conserved.

04 / DISCRETISATION

$$r = c \Delta t / \Delta x, \qquad 0 < r \leq 1,$$
$$u_j^{n+1} = 2u_j^n - u_j^{n-1} + r^2 (u_{j-1}^n - 2u_j^n + u_{j+1}^n),$$
$$j = 1, \dots, N - 1, \quad n \geq 1,$$
$$u_j^1 = u_j^0 + \Delta t \, g_j + \tfrac{1}{2} r^2 (u_{j-1}^0 - 2u_j^0 + u_{j+1}^0),$$
$$g_j = u_t(x_j, 0) = 0, \qquad u_0^n = u_N^n = 0,$$
$$\text{error} = O(\Delta t^2 + \Delta x^2) \quad \text{for smooth solutions.}$$

A standing wave exchanges kinetic and strain energy. Its total energy remains constant.

Select and double-click any equation or diagram to edit its source. Scroll to work below.